CHAPTER 7

RULES FOR MOVEMENT AND BLOCK WORKING

701 General

(1) **Categorization of Track Machines -** For the purpose of movement and working, track machines are classified in two categories

(a) **Track Machines Hauled by Locomotive -** TLE, TRT, PCCM, MDU are the machines generally falling in this category. Rules for movement for material trains will apply to these categories of machines. Details of movement and working are elaborated in this chapter. Other track machines when attached with the train/locomotive will also fall in the same category.

(b) **Self-propelled Track Machines -** Rules for movement and working of self-propelled track machines are detailed in this chapter. This category of track machines may also work with the non-self-propelled track machine as in (a) above for effective block utilization, in the same block section.

(2) **Self-propelled Track Machines -** For movement, these shall be worked as a train but will not need a guard or a brake van. One camping coach or one BRN wagon or another self-propelled track machine, which may be un-braked, shall permitted to be hauled by the self-propelled track machine. SSE/JE/TM operating the machine shall ensure safe running.

(3) **Provision of G&SR -** Provisions as contained in GR 4.65 supplemented by SR framed by the Zonal Railways, shall be applicable for working of track machines. Presently, SRs of Zonal Railways are not uniform. Objective should be to have common or unified SR for all Zonal Railways for common aspects of track machines working. Specific provisions in respective SRs shall deal with special circumstances prevailing in individual Zonal Railways.

(4) **Knowledge of G&SR -** Track machine staff shall be conversant with the relevant provisions of G&SR. Adequate knowledge of provisions of G&SR shall be imparted during the initial and refresher training courses conducted in ZRTIs. The list of standard forms is given in **Annexure7.1**.

(5) **Manning of Track Machines and Possession of Safety Equipment -** All track machines shall be manned by a competent railway servant authorized to work the machine. These machines shall also be equipped with prescribed set of safety equipment to ensure safe movement and working.

(6) **Salient Information of Track Machine -** Following information shall be painted on the track machine at pre-decided locations:

(a) Designation of Controlling Officer (Dy.CE/TM/Line).

(b) Name of owning Railway.

(c) Track Machine No. with model.

(d) Transportation Code.

(e) Month and Year of commissioning.

(f) Date of last POH and last IOH done.

(g) Maximum permissible speed when self-propelled, coupled with other track machine and when attached with train formation as per speed certificate of RDSO sanctioned by Railway Board.

(h) Maximum height and width of track machine in closed condition.

(i) Axle load, wheel base and overall length of machine.

(j) Jacking point i.e. lift here.

702 Operation and Working of Track Machine

(1) **Responsibility of Track Machines -** For the purpose of operation, each machine shall be under the direct charge of SSE/JE/TM. He shall be in-charge of the track machine and shall be responsible for safe movement and working of track machine. All SSE/JE/TM posted on the machine shall have the knowledge of movement and working of track machines and shall possess valid competency certificates. Total number of persons including railway staff, allowed in the driving cab of each machine, shall not normally exceed five.

(2) **Supervision of Work of Track Machine(s) -** The track machine(s) shall work under the direct supervision of an engineering official, not below the rank of SSE/JE (P.Way), who will be responsible for taking the traffic block, coordinating for shunting in the yard, entry of machine(s) into the block section, protection of the line while the work is in progress and timely clearance of block and track machines from the section to the station, after the completion of the work, ensuring the safety of the track(s). He shall also be responsible for coordination with divisional officials for clearance of section in event of breakdown of machine. SSE/JE (P.Way) shall also associate himself and co-ordinate with station master for shunting of the track machine(s).

(3) Working - General

(a) The machine staff and P. Way staff shall work as a team towards the common goal of ensuring optimum utilization of the machine and manpower.

(b) The machine staff shall ensure adequate attention to the machine in time as prescribed in RDSO and manufacturer's manuals in respect of their machines and keep the machines ready for availing stipulated blocks. They should, in consultation with the P. way staff/Engineering Controller/Track Machine Control, be aware of corridor block/MOU provisions and line block spell and plan out their maintenance activities in such a manner that blocks are availed without any lapse. The machine

in-charge shall invariably inform SSE/JE (P.Way) and if not available, to SM or Engineering Control about readiness or otherwise of the machine for working every day well in time.

(c) The machine shall be berthed in the sidings safely as stipulated in the G&SR and the P.Way staff shall arrange for watchman for the machine during the non-working shift. The watchman for the machine should not be frequently changed.

(d) The machine in-charge on arrival shall check up the condition of the machine and report any unusual features observed by him such as disturbance to the machine at the berthing place, missing of parts etc. and initiate appropriate action as per extant rules.

(e) The machine in-charge and SSE/JE (P.Way) shall jointly inspect and finalize the week's work in advance and discuss the day's programme and share with each other all information required for the working.

(f) SSE/JE (P.Way) shall ensure availability of requisite men and tools & plant to work with the machine and for rendering possible assistance to clear the section in case of breakdowns.

(g) SSE/JE (P. Way) shall be responsible for arranging necessary lighting etc. for night works and also coordinate with other divisional staff such as Signal/OHE etc.

(h) It should be ensured by the machine in-charge that no person climbs to the top of any machine without OHE block in electrified sections.

(i) The machine in-charge shall ensure that all precautionary measures are taken for safety of the staff while working on double/multiple line block section against the danger of trains moving on adjacent lines. SSE/JE (P. Way) should provide lookout-men at the site of machine working as per requirements, including protection as per IRPWM and GR 6.03.

(j) SE/JE (P Way), after completion of work shall inspect the worksite attended by track machine thoroughly and ensure track fitness for running of trains with or without speed restriction, as appropriate. He shall ensure that no infringement, obstructions have been caused to safe train running during working of track machine.

(k) SM will liaison with Section Controller/Controller to ensure that at the end of work, the machine is brought to the base station and placed in the nominated berthing line at the earliest to enable post block maintenance of the machine being undertaken. SSE/JE (P.Way) shall also provide necessary assistance as required.

(l) Due to non-availability of path etc., it may not be possible to bring the machine back to the siding where camping coach of machine is stabled, in some of the busy sections. In such eventuality, Sr. DEN/DEN/AEN shall have the powers and provision to hire the road vehicle to bring the staff

back to machine stabling station.

703 Competencies of Track Machine Staff - SSE/JE/TM shall be competent to work the machine and he should possess adequate knowledge, which would be imparted during initial and refresher courses. Competent authority shall duly certify him for his competence. Each SSE/JE/TM shall have a diary as per proforma given in **Annexure 7.2** of this manual and the details of these competencies/fitness shall be recorded therein by SSE/TM/DI or SSE/TM/SDI, as per the system on Railway; in addition to other relevant information. Road learning will be recorded by LI or nominated supervisor.

(1) Medical Fitness and PME - SSE/JE/TM shall be fit in A-3 medical category. If wearing spectacles, he shall carry one extra pair while on duty. Technicians and helpers working on track Machines shall be fit in B-1medical category. This shall be recorded in certificate of medical fitness at the time of joining and certificate of periodical medical examination (PME) during the course of service. The employee will inform his immediate superior 60 days in advance of expiry of his medical certificate. Employee will be expeditiously booked for PME.

(2) Technical Training - All SSE/JE/TMs are to undergo initial as well as refresher technical training imparted by IRTMTC, Allahabad; successfully. Refresher technical training shall be done every 3 years. Certificate of satisfactory completion of training should be kept in the personal custody of the official and produced when required.

(3) G & SR Training - All SSE/JE/TMs are to undergo initial as well as refresher training in G&SR; successfully. Refresher G & SR training shall be done every 3 years. Certificate of satisfactory completion of training should be kept in the personal custody of the official and produced when required.

(4) Route/Road Learning - It will be the responsibility of the SSE/JE/TM working the machine to acquaint himself with the system of working, location of signals and other local conditions affecting running of trains on a section(s) of the railway over which he is to work and if he is not so acquainted with any portion of the railway over which he is to work, obtain the services of qualified railway servant who is conversant with it to assist him by requesting the office of Dy.CE/TM Line.

SSE/JE/TM will make trips in the cab of locomotive of a train for initial road learning for acquainting with the section as well as subsequent, if required, for reacquainting with the section as stipulated in G&SR of the railway. Entry to this effect shall be made in the diary by the employee and countersigned by LI or SSE/TM/SDI of concerned satellite depot. For this purpose, posting of at least one LI in each division, depending on the number of machines i.e. number of SSE/JE/TM, shall be ensured.

(5) Competency Certificate - SSE/JE/TM shall be considered competent to operate the machine only if he possesses valid machine competency

certificate. Machine competency certificate is to be issued to SSE/JE/TM by Dy.CE/TM Line or an officer authorized by him. This certificate will be issued as per proforma given in **Annexure 7.3** after ascertaining the successful completion of technical training, G & SR training and his medical fitness. The validity of this certificate will be up to the earliest expiry date of the three i.e. (i) Technical training (ii) G & SR training and (iii) PME.

For automatic block section, separate competency is required to be issued as per the practice in the Zonal Railway.

704 Safety Equipment

(1) **General** - SSE/JE/TM in-charge shall be responsible to ensure that the following equipment in working condition are available on the track machine:

(a) Two red and one green hand signal flags.

(b) Two tri-colour hand signal lamps /LED torch.

(c) Two chains with padlocks.

(d) One fire extinguisher in each cabin.

(e) Two hooters (manually controlled).

(f) Two jacks10 t.

(g) Four wooden blocks.

(h) Four crow bars.

(i) One hydraulic hand pump.

(j) Emergency pneumatic/hydraulic hose off sizes suiting to different machines (Complete with end fitting).

(k) Wire rope with close loops at both ends 2 m and 9 m long for BCM: One of each length.

(l) Machine specific equipment, if any, listed in Chapter 2, 3, 4 and 5.

(m) Ten fog signals (detonators) in a tin case.

(n) A copy of the working timetable of the section where the machine is working.

(o) G & SR book with up to date amendment slips.

(p) One 4 cell flasher light LED lamp cum flasher light (rechargeable).

(q) Two banner flags.

(r) One first aid box.

(s) Two skids.

(t) Safety helmets for all machine staff.

(u) Protective clothing, safety shoes and safety gloves.

(v) Walkie talkie with frequency of SM, Guard and Loco Pilots.

(w) Internal communication system like walkie-talkie and/or head mounted system.

(x) Track Machine Manual with up to date correction slips.

(y) Accident Manual.

(z) Tail lamp.

(2) **Head and Tail Lights –** Each track machine must be equipped with prescribed head and tail lights, marker lights and flasher lights as per GR 4.14, 4.15 & 4.16 and SRs thereof. Each machine shall display LV board/tail lamp when moving alone. While moving in conveyor coupled, the LV board/tail lamp shall be fixed on the last vehicle; in the direction of movement.

705 Rules for Operation – General

(1) **Stabling of Track Machines -** When the track machine(s) is/are stabled at a station, SSE/JE/TM in-charge shall ensure that it is clear of fouling marks and traps and necessary precautions against rolling down such as pinning down hand brakes, chaining and provision of skids; is taken in accordance with G&SR.

(2) **Shunting of Track Machines -** No track machine shall be moved between a running line and the siding/stabling line without the written permission of the Station Master on duty in the form of shunting order/shunt signals.

(3) **Machine Ready Memo -** SSE/JE/TM shall issue a written machine ready memo (as per Annexure 7.4) after necessary maintenance/repairs/schedules and Brake Power testing and other stipulated checks, if any, to on duty SM, indicating time and date, under advice to SSE/JE/P.Way deputed to work with the machine.

(4) **Movement of Track Machines -** When the track machine is required to move from one station to another station, SSE/JE/TM shall run the machine only with proper authority to proceed and all necessary permits, notices and cautions as specified in G&SR. When track machine is to move on wrong road (against the direction of traffic), the speed of track machine shall not exceed more than 25 kmph and flasher light shall be kept “ON”.

(5) **Working in Group**

(a) When more than one machine is required to work within the same block section, these machines may be allowed to move into the block section in a group under one authority as detailed in this chapter. In such situation all the track machines must leave and enter the section simultaneously one after another keeping adequate distance among them and with proper authority as detailed further in the following paras.

(b) Total number of the machines shall be clearly mentioned in the line clear/block authority message with exchange of private numbers. For

this purpose, two coupled self-propelled track machines are to be treated as single machine. Track machine with one BFR, like UTV, shall also be treated as a single machine.

(c) In case of group machines, entering into the block section for working, and back and entering the station after working, the first machine shall pass the dispatch/reception/last stop signal, as the case may be, in OFF position (where signal is given) and all other machines shall be piloted by competent railway servant following first machine with signal in ON position. In case where the machines are allowed on paper authority instead of signals, SSE/JE/P.Way or SSE/JE/TM in the first machine will receive the authority and other machines will follow duly piloted by competent railway servant.

(d) Normally machines shall not work in a group while visibility of track is restricted due to sharp curves and steep gradients. Where the working of track machines in group is indispensable i.e. like deep screening, track relaying site etc., the necessary block protection shall be done as per para 806 of IRPWM and special precaution shall be taken as per para 811 of IRPWM for the first machine facing the direction of traffic in double line section and for the front and rear most machines in single line section.

(6) Run through movement in Group - Run through movement of the track machines in-group is not allowed. They can however transit as coupled vehicle under one authority. However, the machines can move in group by taking a short duration block and rules as for group movement for block shall be followed.

(7) Working of Track Machine in Block

(a) The track machine(s) shall work under line block only. The SSE/JE (P. Way) shall give the requisition for block in duplicate to Station Master on duty, as per proforma in **Annexure 7.5,** indicating therein specific location where the machine(s) will work, the road i.e. UP/DN etc., the number of track machine(s) that will work along with individual number(s) and sequence (order in which machines will leave the station), the duration of block required and whether the machine(s) will proceed to the next station or return back to the starting station after the work. The SM shall then contact the control and ascertain the movement of trains before granting line block.

(b) After the block is granted, the Station Master will return the original copy of requisition endorsing the duration and other special instructions like work and proceed, work and return via right or wrong direction, caution order, if any SSE/JE (P. Way) shall accompany the machine(s) to the work site.

(c) SSE/JE (P.Way), shall get the requisition so returned by station master

(also called as track machine permit), signed from the SSE/JE/TM in-charges of track machines.

(d) On reaching the station as mentioned in track machine permit, after the completion of work, SSE/JE (P.Way) will hand over the token, track machine permit and the key etc. if any, after all the machine have cleared the block section. He will also certify that the track is fit for train movement. Then only the Station Master will clear the line for normal train operation subject to the observations of speed restriction, if any.

(8) Working in Automatic Block Sections – Automatic signalling territory involves special features in working and the special instructions for working of track machine are elaborated in **Annexure 7.6**

(9) Working in Integrated Traffic Blocks - If the track machines are working in an integrated traffic block where other machines and staff engaged by other departments are also working, the relevant instructions issued by Zonal Railway for integrated block working should be followed.

(10) Use of Cell Phone on Track Machine - All supervisory staff in the track machine and as well as at the track machine site is prohibited from use of mobile phones during movement and working of track machine, except in emergencies or when it is extremely necessary.

(11) Requirement of Pilot - If the SSE/JE/TM is not acquainted with the section, where the track machine has to work, a Pilot (Loco Pilot/ALP) shall be provided for movement as well as block working as per the specific request in advance. SSE/JE (P.Way) may also work as pilot in his jurisdiction, in case of emergent situations.

(12) SSE/JE(P.way) as Guard - In emergent and special situations, SSE/JE/P.Way shall work as the guard. SSE/JE/P.Way shall perform the limited functions of a guard required for working and movement of UTV, which has only one SSE/JE; as it has one BFR and direct visibility of points from the driving cabin may not be there in one direction.

706 Working Of Track Machines In Single and Double Line Sections

(1) Single Line Section – Work and Proceed

(a) Station Master will obtain line clear from station in advance, take off last stop signal, issue track machine permit and handover the token, if any, to SSE/JE (P.Way), SSE/JE (P.Way) shall travel in the last track machine.

(b) On completion of the work, the machine(s) will be received by taking off reception signals. Competent railway servant should display green hand signal at the foot of first stop signal till the last machine enters the station.

(c) Station Master will clear the line for normal train operation only after SSE/JE (P.Way) has returned the track machine permit etc. and certified

the track fit for train movement.

(2) Single Line Section – Work and Return

(a) With token / tablet instruments

(i) Station Master will obtain line clear from station in advance; take off last stop signal, issue track machine permit along with token/tablet to SSE/JE (P.Way). SSE/JE (P. Way) shall travel in the first track machine.

(ii) On completion of the work, the machine(s) will be received by taking off reception signals. Competent railway servant shall display green flag/signals at the foot of first stop signal till the last machine enters the station.

(iii) Station Master will clear the line for normal train operation only after SSE/JE (P.Way) has returned the track machine permit etc. and certified the track fit for train movement.

(b) With token less Block Instrument

(i) Station Master will block back the section, take out the shunting key, issue track machine permit, and shunting key to SSE/JE (P.Way) In addition requisite written authority will be issued for passing the last stop signal at danger for all machines SSE/JE (P.Way) shall travel in the first track machine.

(ii) On completion of the work, the machine(s) will be received by taking off reception signals. Competent railway servant shall display green hand signals at the foot of first stop signal till the last machine enters the station.

(iii) Station Master will clear the line for normal train operation only after SSE/JE (P.Way) has returned the track machine permit etc. and certified the track fit for train movement.

(3) Double Line Section – Work and Proceed

(a) Via Right Direction

(i) Station Master will obtain line clear from station in advance, take off last stop signal, issue track machine permit SSE/JE (P.Way) shall travel on the last track machine.

(ii) On completion of the work, the machines will be received by taking off reception signals; competent railway servant shall display green flag/signals at the foot of first stop signal till the last machine enters the station.

(iii) Station Master will clear the line for normal train operation only after SSE/JE (P.Way) has returned the track machine permit etc. and certified the track fit for train movement.

(b) Via Wrong Direction

(i) Station Master will take the line clear from station in rear on block telephone, prepare paper line clear ticket, issue track machine permit, to SSE/JE (P.Way), SSE/JE (P. Way) shall travel on the last machine.

(ii) The machine(s) shall be piloted out of the station on a written authority issued by the Station Master after all the facing points have been correctly set and locked and trailing points correctly set over which the machine(s) will pass.

(iii) On approaching the next station after completion of the work, SSE/JE/TM shall bring their machines to stop opposite the first stop signal pertaining to the right line or at the last stop signal pertaining to the wrong line (on which they are running) whichever they come across first and wait and proceed as piloted or follow GR 4.44.

(iv) The Station Master at the other end of the block section shall depute a railway staff in uniform at the foot of the signal (whichever the machines would encounter first) who shall stop the machines on danger hand signal and thereafter pilot them into the station on a written authority issued by the Station Master.

(v) If the operators find that no railway servant in uniform has been deputed at the foot of the signal to pilot the machine(s) into the station, GR 4.44 shall be observed.

(vi) All the crossover points in the facing direction, over which the machines shall proceed, shall be clamped and pad locked. This fact will be mentioned in the authority in prescribed form issued by station master, through the competent railway servant.

(vii) Station Master will clear the line for normal train operation only after SSE/JE (P.Way) has returned the paper line clear ticket, track machine permit etc. and certified the track fit for train movement.

(4) Double Line Section - Work and Return

(a) Via Right Direction

(i) Station Master will block forward the section and arrange to put the needle of the block instrument directly to "Train on Line", take out the shunting key in case of 'Daido' Double Line Block Instrument, issue track machine permit along with the shunting key, if any, to SSE/JE (P.Way). In addition, requisite written authority will be issued for passing the last stop signal at danger. SSE/JE (P. Way) shall travel in the first track machine.

(ii) On completion of the work, SSE/JE/TM shall bring their machine to stop opposite first stop signal pertaining to the right line or at the

last stop signal pertaining to the wrong line, whichever they comes across first.

(iii) Station Master shall depute a railway servant in uniform at the foot of the signal (whichever the machine would encounter first) who shall stop the machine(s) on danger signals and thereafter pilot them into the station on a written authority issued by the Station Master.

(iv) If SSE/JE/TM find that no railway servant has been deputed to pilot the machine(s), GR 4.44 shall be observed.

(v) All the crossover points in the facing direction over which the machines shall proceed, shall be clamped and pad locked.

(vi) Station Master will clear the line for normal train operation only after SSE/JE/P.Way has returned the track machine permit etc. and certified the track fit for train movement.

(b) Via Wrong Direction

(i) Station Master will block back the section as per extant instructions, put the needle of the block instrument directly to "Train on Line" take out the shunting key in case of 'Daido' Double Line Block Instrument issue track machine permit special caution order indicating the number of track machine(s), station where they will return etc. which will be signed by the operators and then will be handed over along with the shunting key, if any to SSE/JE (P.Way). SSE/JE (P.Way) shall travel in the first track machine.

(ii) The machines shall be piloted out of the station on a written authority issued by the Station Master after all the facing points have been correctly set and locked and trailing points correctly set over which the machines will pass.

(iii) On completion of the work the machines will be received by lowering reception signals. Competent railway servant shall display green flag/hand signal at the foot of first stop signal till the last machine enters the station.

(iv) Station Master will clear the line for normal train operation only after SSE/JE (P.Way) has returned the track machine permit etc. and certified the track fit for train movement.

707 Important Instructions and Precautions

(1) Protection of Work Site - SSE/JE (P.Way) shall be responsible for protection of the site of work and also protection of adjoining track(s) in case of infringement. He shall be conversant with the infringing conditions of the various machines. He shall also be responsible for safe condition of track before clearing the block after machine working.

(2) **Fouling of Adjacent Lines -** Some machines like BRM, PCCM etc. tend to foul the adjacent line, while working on double line section or in yard. If any part of a machine is likely to foul the adjacent line(s) while working, SSE/JE (P.Way) shall request Station Master in writing to block such line(s) likely to infringe. Such work shall only be undertaken if blocking of such adjoining line(s) has been permitted by the control and the Station Master and such adjoining lines have been protected. Efforts shall be made to ensure the blocking of other lines for minimum duration.

(3) **Information to Level Crossings -** While exchanging private numbers with level crossings, the Station Master on either side shall inform all the level crossings equipped with telephones falling in the block section.

(4) **Safe Distance and Speed of Machine in Group -** The minimum distance between the machines when working in a group shall be 50 m to avoid collision between machines and danger to life of machine and P.Way staff working with machine. While the track machines are moving in the block section in group, it will be the responsibility of SSE/JE/TM of these machines to maintain a minimum safe distance of 200 m from each other. The speed of movement of track machines shall be the lowest of the permissible speed among all track machines. The leading track machine shall observe this speed. If any of the machines is required to slowdown or stop due to some reason, SSE/JE/TM driving the machine should ensure that red hand signal is displayed by waving vigorously. Where visibility is poor such as on curves and in cuttings, appropriate speeds and safe distance should be maintained to ensure safety. While approaching the level crossings, SSE/JE/TM driving the track machines shall keep a vigil for any obstruction and whistle freely till the machine passes the level crossing.

(5) **Setting of Points of Stabling Siding -** The concerned points of non-signalled sidings shall be set against the line on which the track machine is stabled and such points shall be secured with clamps or cotter bolts and pad locks as stipulated in G&SR. The keys of such padlocks shall be kept in the personal custody of Station Master until the machine is ready to leave from siding or running line. SSE/JE/TM in-charge shall not relinquish charge until he has satisfied himself that the machine has been properly secured and protected as prescribed herein.

(6) **Shunting in Line occupied by Track Machine -** During shunting on a line occupied by track machines, station master shall ensure that no machine is shunted without the presence of SSE/JE/TM.

(7) Safety of Men

(a) Working of certain track machines creates conditions jeopardizing safety of man and material, if adequate precautions are not enforced at site of work. BCM is likely to produce a dusty atmosphere impairing visibility in the vicinity of machine and also heavy noise. TLE and TRT

involve large number of labour working with the machine. Hence, extra care is necessary as detailed below, to ensure safety of workers.

(b) Hooters should be provided on the track machines. These hooters should be used to warn the staff working on/around the track machine about approaching train on adjoining track. Remote controlled hooters shall also be deployed as an added precaution by SSE/JE/P.Way so that lookout man standing around 150 m away from the track machine can also operate the hooter to warn the staff suitably. SSE/JE/TM shall also put on the flasher light on as an added precaution till the train on adjacent line has passed the site of work.

(c) Caution order of 30 to 50 kmph with instructions to whistle freely should be imposed on the adjacent line, during the duration of block, for the safety of workmen, depending upon the site conditions and visibility.

(8) **Checking Infringement After Work -** The vertical and lateral clearance for OHE, signal post and any other structure should be checked and adjusted before clearing the block. It shall be ensured by SSE/JE (P.Way) working with track machine that there should be no infringement to signal post, OHE and any other structure as per schedule of dimensions.

708 Failure and Accidents of Track Machines

(1) **Protection in case of Breakdown -** In the event of breakdown, the track machines shall be protected as per GR 6.03 and SR there to by the machine staff, as directed by machine in-charge.

(2) **Failures in Block Section -** Failures in block sections of the track machines will be treated as accident under class 'J – Equipment failure'.

(3) **Accidents involving Track Machine -** Accidents involving track machines shall be treated as train accidents under the appropriate class and action shall be taken as per the rules in force.

(4) **Action in case of Failure in Block -** In case of failure of track machine in block section, immediate information with details should be conveyed to the ADEN/DEN/Sr.DEN of the section and the AXEN/XEN/Dy.CE/Line/TM. SE/JE/TM should decide in consultation with SSE/JE (P. Way), the action to be taken to clear the section. They may decide to push the disabled unit to the nearest station provided the brake power is in good condition. Otherwise, intimation shall be sent to the nearest Station Master asking for a light engine to tow the unit.

(5) **Request for ART/Breakdown -** In case, SSE/JE (P. Way) and/or SSE/JE/TM feels clearance of section is going to take long time, the assistance of Road Break-down or Accident Relief Train shall be asked for immediately. Meanwhile SSE/JE/TM in-charge on the machine shall take necessary action to rectify the defect(s). SSE/JE (P. Way) shall provide all necessary assistance.

709 Permissible Speed - Each machine will run within the maximum permissible speed sanctioned for that type of machine on a given section of the Railway.

710 Speed Certification for Track Machines - Railway Board Policy Circular No. 6 governs procedure for certification of permissible speed for new rolling stock. The provisions of the circular relevant to track machines are summarized below:

(1) Speed Certification for New type of Track Machine for Regular Movement

Any new type and model of track machine introduced on Indian Railway is allowed to run on Zonal Railways at a speed specified in the provisional speed certificate issued by RDSO. This can be done only after the approval of Railway Board processed through Chief Commissioner of Railway safety on the application processed by the RDSO (Gazette notification no.945(E)).

Final speed certificate for the track machine is also issued by RDSO, after conducting detail oscillation trials. For conducting trial, on the advice of RDSO to test track machine, General Manager of the concerned railway shall allow conducting the trials on their system on the test section identified by RDSO, after obtaining permission from CRS in case trial is to be done on passenger running lines with Joint Safety Certificate duly signed by the Principal Chief Engineer, Principal Chief Mechanical Engineer, Principal Chief Operating Manager and Principal Chief Signal & Telecommunication Engineer (Principal Chief Electrical Engineer also in case of locomotive or involvement of electrified section).

GM sanction for such trials shall be valid for two years, after which it shall require revalidation by the General Manager on the advice of RDSO. With regard to a stock already operating on provisional speed certificate, while revalidating the permission for oscillation trial, any adverse performance or special observations made during service, if any, shall be commented upon.

Sanction of Railway Board for final maximum permissible speed through Chief Commissioner of Railway safety (CCRS) is also required to be obtained in a similar manner as above. The procedure to be followed is detailed below:

(a) Determination of Provisional Maximum Permissible Speed/ Final Maximum Permissible Speed by RDSO

(i) **Provisional maximum permissible speed**

- The provisional maximum permissible speed for new type of track machine shall be determined on the basis of design features and other details supplied by the OEM. Guidance can also be taken from the permissible speed of similar type of machine already in service.
- The provisional speed will normally be lower than the design or projected maximum permissible speed but shall not be more than 65 kmph. However, for track machines, whose suspension and other relevant characteristics such as axle

load, track loading density etc. are similar to stock already in service, maximum permissible speed may be increased to 75 kmph by Executive Director Standards (Motive Power)/RDSO in consultation with Executive Director Standards (Track) and Executive Director (Bridges & Structure) and other concerned Directorates.

- The provisional speed certificate will be valid for five years except when it is superseded by final maximum permissible speed certificate issued following the procedures described below:

(ii) **Final maximum permissible speed**

- The final maximum permissible speed shall be determined on the basis of detailed oscillation trials for assessing the riding quality and/or stability. However, for such stock whose suspension and other relevant characteristics such as axle load, track loading density etc. are not basically different from those of the existing ones, the detailed oscillation trial may be dispensed by Executive Director Standards (Motive Power)/RDSO in consultation with Executive Director Standards (Track) and Executive Director (Bridges & Structure) and other concerned Directorates. In case of difference of opinion, the matter shall be referred to the Railway Board for final order.
- For conducting oscillation trials, a separate provisional speed certificate for oscillation trials, shall be issued by RDSO specifying the section where trial is to be conducted.
- General Manager of the concerned railway shall allow conducting the trials on their system on the test section identified by RDSO, after obtaining permission from CRS in case trial is to be done on passenger running lines with Joint Safety Certificate duly signed by the Principal Chief Engineer, Principal Chief Mechanical Engineer, Principal Chief Operating Manager and Principal Chief Signal & Telecommunication Engineer (Principal Chief Electrical Engineer also in case of locomotive or involvement of electrified section).
- Oscillation trial is conducted under the supervision of by RDSO.
- Final maximum permissible speed certificate shall be issued by RDSO based on the results of detail oscillation trial conducted.

(b) Application for Sanction by RDSO for introduction of Machine

1) The RDSO shall apply, for introduction of machine different from those already running on any section or division of zonal railway for use by any zonal railway to the Railway Board through the Chief Commissioner of Railway safety (CCRS).

   Any increase in sanctioned maximum speed of machine shall be treated as introduction of new types of locomotives or rolling stock.

2) The application shall be accompanied by-

   a) such diagrams as may be necessary to give full particulars of the axle loads, wheel spacing, length over buffers and other principal dimensions of the machine for which sanction is required.

   b) provisional speed certificate or final speed certificate, as the case may be, issued by the RDSO.

3) The application shall be scrutinized by the Commissioner and his recommendations thereon shall be submitted to the Railway Board for its orders.

4) A machine have different principal dimension or a different bogie design or new designs of braking system or suspension details like axle load, track loading density, unsprung mass being different shall be regarded as new machine.

   Provided that any minor change of equipment design or change of internal equipment layout on the machine or minor change of axle load or minor change of track loading density or minor change in unsprung mass shall not be regarded as new machine, unless such changes are likely to significantly affect weight distribution, centre of gravity or riding behaviour of the machine.

5) In case of any difference of opinion in the RDSO, whether or not any change or modification is to be regarded as a new rolling stock, the RDSO shall, through the Commissioner, refer the matter to the Railway Board for final decision thereon.

(c) Railway Board's Sanction for introduction of Machine – Railway Board grants sanction for provisional introduction of machine at Provisional/Final speed on the basis of recommendation of Chief Commissioner of Railway Safety.

(2) Use of Machine by Zonal Railways

(1) The General Manager of Zonal Railway may sanction the use of machine already introduced on Indian Railway by the Railway Board on any section or division of railway under his control.

(2) The proposal for sanction of the General Manager of a Zonal Railway accompanied by-

(i) Such diagrams as may be necessary to give full particulars of the axle loads, wheel spacing, length over buffers and other principal dimensions of the rolling stock for which sanction is required.

(ii) the provisional speed certificate or final speed certificate, as the case may be, issued by the RDSO.

(iii) the sanction of the Railway Board for introduction of the machine or increase in sanctioned maximum speed of machine.

(iv) the calculations and stress sheets showing-

a) the conclusions arrived at.

b) the external forces on which the stress calculations are based.

c) the stresses which will be produced in the various bridges over which the proposed rolling stock will run.

d) the effects which the said rolling stock will have on various structures or tracks as compared with those caused by the machine already in use, or allowed by the existing orders.

Provided that the calculations and stress sheets under this clause must show as to what allowance has been made for any secondary or deformation stresses in addition to the primary stresses caused by external forces and what relief of stress, if any, has been included.

(v) the cost of modification to signalling and telecommunication installations necessitated by the use, if any, of chopper or thyristor control systems.

(vi) an approximate estimate of the cost of such improvements in existing structures or track as the use of the proposed rolling stock is likely to render necessary on the concerned railway, whether immediately or in the near future.

(vii) a certificate signed by the Chief Engineer, the Chief Mechanical Engineer and the Chief Electrical Engineer (for electric stock) of the concerned railway in the following proforma, namely-

CERTIFICATE

Certified that it is safe to run----------(particulars of locomotive or rolling stock proposed to run)not exceeding----------units(in the case of locomotive) coupled together on the section---------(station) to ----------(station) from ----------(km) to ----------(km) of the ---------- Railway at a maximum speed of ----------(km/h) against a maximum speed of ----------(km/h) certified by Research Design and Standards Organization, subject to the following speed restrictions and conditions;-

(a) Speed restrictions: -

Sl. No.	From Km to Km	Nature of speed restriction	Brief Reason for restriction

(b) Special Conditions-

1)……

2) ……

3) ……

4) ……

To be signed by-

1. The Chief Mechanical Engineer……. 2. The Chief Engineer…….

3. The Chief Electrical Engineer……. 4. The Chief Operation Manager…….

5. The Chief Signalling and Telecommunications Engineer…….

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(3) The certificate referred to shall indicate-

(a) Clearly that the speed certified does not exceed the limits laid down by the Research Designs and Standard Organization; and

(4) The General Manager of the zonal railway shall communicate the sanction to the Commissioner for his information ten days in advance of the actual use of machine over the railway.

(3) One Time Movement of New Machine

a) The maximum permissible speed for the limited purpose of movement of machine from the manufacturer's works/docks to destination or to the

testing point or from the destination/testing point back to manufacturer's works shall be determined and certified by Executive Director Standards (Motive Power) in consultation with Executive Director Standards (Track) and Executive Director(Bridges & Structures) and other concerned Directorates. The speed for this purpose shall not be higher than the provisional speed determined by RDSO.

b) The maximum permissible speed prescribed by the RDSO shall be subject to approval by the Principal Chief Engineer and Principal Chief Mechanical Engineer (Principal Chief Electrical Engineer also in case of locomotive or involvement of electrified section) of the concerned Zonal Railways in the form of JSC, who shall ensure that the track, bridges and OHE in the sections concerned are suitable for the new stock at the speed permitted. However, if a new machine infringes the Indian Railway Schedule of Dimensions or axle loads are more than that permitted on the section; JSC duly signed by Principal Chief Engineer, Principal Chief Signal & Telecom Engineer and Principal Chief Mechanical Engineer (Principal Chief Electrical Engineer also in case of locomotive or involvement of electrified section) of the concerned Zonal Railways, should deal these issues and prior approval of GM shall be taken.

c) In such cases, Commissioner of Railway Safety shall be kept informed for the movement, but, no formal approval of the Commissioner of Railway Safety is essential. However, in case it becomes necessary to move the vehicle attached to a passenger carrying train, sanction of the Commissioner of Railway Safety shall be taken.

d) For each single movement of any machine not covered above, joint safety certificate shall be signed by Principal Chief Engineer, Principal Chief Mechanical Engineer, Principal Chief Operating Manager and Principal Chief Signal & Telecommunication Engineer (Principal Chief Electrical Engineer also in case of locomotive or involvement of electrified section) and submitted to CRS for his permission. CRS while permitting such movement may impose any special stipulations as deemed necessary for such movement.

711 Infringements - The Railway Board is the final safety controlling authority for the Indian Railways and it is the final authority for regulating and sanctioning speeds of all rolling stocks under the Indian Railways Act. For permitting any vehicle on Indian Railways, if any of the dimensions of machines is infringing the Indian Railway Schedule of Dimension, it shall have to be first condoned by Railway Board. The application for such condonation shall be initiated by RDSO through Chief Commissioner of Railway Safety.

Details of such condonations for older machines condoned under **Schedule of Dimension– 1939** and new machines infringing **Schedule of Dimensions –2004** are given in **Annexure -7.7**, machine-wise

712 Special Precautions while working in Special Circumstances

(1) Working of Machines in Ghat Section - The following precautions are essential for safe driving of the machine in sections involving steep gradients, sharp curves, deep cuttings high embankments etc.

(a) Ghat Section – means " A part of block section between any station of Indian Railway route which consists of 70 % in length or more that goes through hilly areas with significant continuous gradients i.e. 1 in 100 or steeper shall be called as Ghat section".

(b) SSE/JE/TM shall have Block Competency Certificate for ghat section working or a pilot driver having knowledge of ghat section where track machine(s) has to work, shall be deputed on track machine(s).

(c) Before offering readiness, brake power, hand brake & emergency brake shall be tested.

(d) SSE/JE/TM shall not leave driving/working cabin of track machine in ghat section.

(e) Wheel skids shall be kept ready.

(f) While going in steep down gradient, a loco shall be attached/coupled in front/leading direction of track machine and loco shall control all movements.

(g) While going in steep up gradient, a loco shall be attached/ coupled in rear/trailing direction of track machine direction and loco shall control all movements.

(h) For working purpose, track machine will be uncoupled & loco shall move along 25 meters away from the track machine. After completion of work, track machine shall be coupled with banking locomotive/banker.

(i) SSE/JE/P.Way shall be responsible to arrange a banking locomotive/banker before readiness of track machine.

(j) Drive only from the cabin corresponding to the direction of travel (except during small shunting movements where movement in opposite direction could be done with adequate precautions).

(k) Do not drive the machine at speeds exceeding the prescribed speed for the section, this speed being further reduced over switches and speed restriction zones.

(l) Never drive over slopes or descending gradients without putting into gear and do not switch off gear box key switch.

(m) Do not let the RPM of the engine fall below 1000 and do not switch off the engine before the machine stops in gradient sections.

(n) Shift back to lower gears well in time in case of ascending/descending

gradients.

(o) Have a special check for the ZF gear box oil level and its working temperature.

(p) Ensure proper working of pneumatic circuit and functioning of all components in the brake system.

(q) Ensure availability of spares particularly in respect of items that are failure prone such as transmission gear like cardan shafts, coupling bolts etc. Availability of a skid is a must.

(r) Also ensure proper working of horns and lights.

(s) Be on the lookout for trespassers while negotiating sharp curves and also sound horns on such occasions.

(t) No run through movement shall be done in convoy while moving in ghat section. Only one machine shall be allowed to move for run through movement in ghat section at a time.

(2) **Night Working -** For machine working in the night suitable lighting arrangements shall be done by SSE/JE/P.Way. Arrangements to take care of movement of lights with machine with progress of work should also be done.

Annexure 7.1

List of standard forms

S.No	Description	Form No	Font Colour /Remarks
1	Starting permit	T 310	Blue/Black
2	Advance authority to pass defective signal	T/ 369 – (1)	Blue
3	Authority to pass signals in 'ON' or defective position	T/369 – (3b)	Blue
4	Authority to pass shunt signal/gate signal at 'ON' and shunting permitted indicator	T. 370	Blue/Black
5	Caution order	T/409	Green
6	'Nil' caution order	T/A 409	Green
7	Reminder caution order	T/B 409	Green
8	Authority to receive train on obstructed line	T/509	Blue
9	Authority to receive train on a non-signaled line	T/510	Manuscript
10	Authority to start a train from non-signaled line	T/511	Blue
11	Authority to start a train from a line with common starter signal	T/512	Blue
12	Authority to proceed for relief engine/train to occupied Block section	T/A602	Red
13	Authority for opening communication during total interruption of communication on single line section	T/B 602	Red
14	Authority for working of trains during total interruption of communication on double line section	T/C 602	Red
15	Authority for temporary single line working on double line section	T/D 602	Red
16	Conditional line clear ticket up	T/G 602	Red

17	Conditional line clear ticket down	T/H 602	Red
18	Shunting order	T/806	Blue
19	Authority to pass automatic/semi automatic/manually operated/gate signals	T/A 912	Blue
20	Authority to proceed without line clear on automatic block signaling territory	T/B 912	Red
21	Authority to proceed relief engine/train into automatic occupied block section	T/C 912	Red
22	Authority to pass automatic block system during prolonged failure of signals	T/D 912	Blue
23	Authority to proceed without Line clear (UP)	T.1408/UP	Blue/Black
24	Authority to proceed without Line clear Down	T.1408/DN	Blue/Black
25	Paper Line clear ticket UP	T/C 1425	Blue
26	Paper line clear ticket DN	T/D 1425	Blue

Annexure 7.2

Personal Diary

Name_____ Designation _____

This book will be in possession of SSE/JE/TM whose photograph is affixed above. He shall produce the same on demand by any inspecting official.

XEN/AXEN/TMC/……

INDEX(1 page desirable)

S.No.	Description(PLEASE CORRECT INDEX)	Page No.
I.	Self Particulars	01
II.	Particulars of Transfers/Promotions	
III.	Particulars of experience on different types of machines	
IV.	Details of Awards/ Achievement's during service period.	
V.	Particulars of IRTMTC Training(Initial/Refresher)	
VI.	Particulars of IRTMTC Training(Special courses)	
VII.	Particular of ZTC Training	
VIII.	Particulars of PME	
IX.	Particulars of Machine Competency	
X.	Particulars of LRD	
XI.	Observation /Remarks of Inspecting officials	
XII.	Observation /Remarks of Inspecting Supervisor	

I. Self Particulars*(1 page desirable)*

S.No	Item Description	Details
1	Name	
2	Father's Name	
3	Date of Birth	
4	Designation and HQ station	
5	Educational qualification	
6	Scale and basic pay	
7	Phone and Mobile No.	
8	Date of appointment to Railway	
9	Date of appointment/Promotion to JE/SSE/TM	
10	Date of Appointment/Promotion to SSE/TM	
11	Unique ID/PF Account No.	
12	Pan No.	
13	Aadhaar Card No.	
14	Bill Unit No	
15	Address	
16	Phone No	
17	Academic Qualification	
18	Blood Group	
19	Date of attaining 45 years of age	
20	Date of attaining 55 years	

II. Particulars of Transfer/Promotion*(1 page desirable)*

SN	Name of the supervisory post	Date of Transfer*/ Promotion	Remarks
01	JE/……		
02	SSE/…….		

***Date of transfer means the date of joining the new post**

III. Particulars of experience on different types of machine (2 pages desirable)

Year@	Machine Type#	Period worked* (months)	Remarks
2016-17			

@ Period taken for filling the details shall be financial year

Machine type is to be written as DOUMATIC, BCM, DGS etc.

*Periods of working on a type of machine of less than 3 months a year are not to be recorded

IV. Details of Award / Achievement's during service life(1 page desirable)

SN	Division Level		Headquarter Level		
	Cash / Certificate	Awarded by	Cash / Certificate	Awarded by	Sign of SSE/TM/SDI

	of age	
21	Family Particulars	
22	Contact No. In case of Emergency	
23	Present Address and home contact No.	
24	Permanent Address and home contact No.	

V. Details of IRTMTC Training (Initial/Refresher Course) (1 page desirable)

Name of Course	Period		Due date for next course	Sign of SSE/TM/SDI
	From	To		

VI. Details of IRTMTC Training (Special Course) ***(1 page desirable)***

Name of Course	Period		Due date for next course	Sign of SSE/TM/SDI
	From	To		

VII. Details of ZTC Training (G&SR)(Initial/Refresher course)(1 page desirable)

Name of Course	Period		Due date for next course	Sign of SSE/TM/SDI
	From	To		

VIII. Particulars of PME *(1 page desirable)*

Category	Date of PME	Certificate No.	Issued by	Next PME Due Date	Sign of SSE/TM/SDI
Up to age of 45 @ 4 year. **For age between 45 to 55 @ 2 year** **Above age of 55 @ every year**					

IX. Particulars of Machine Competency *(1 page desirable)*

(Dy.CE /TM Line or an officer authorized by him)

Date	Validity Period		Issuing Authority	Due date for reissue	Sign of SSE/TM/SDI
	From	To			

X. Particulars of LRD(Road/Route Learning and Signal particulars)

(4 pages desirable)

Section	Line UP/DN etc.	Date of Completion	Sign of JE/SSE/TM	Sign of LI or SSE/TM/SDI

XI. Observations/Remarks of Inspecting Officials*(6 pages desirable)*

Date	Designation	Date of Completion Sign of JE/SSE/TM	Remarks	Sign of LI or SSE/TM/SDI

XII. Observations/Remarks of Inspecting Supervisors *(16 pages desirable)*

Date	Designation	Date of Completion Sign of JE/SSE/TM	Remarks	Sign of LI or SSE/TM/SDI

Annexure 7.3

Competency Certificate

Office of Dy.Chief Engineer,

Track Machine Line,

……, …….

Photograph

This is to certify that Shri…….. Designation…… is competent & authorized to operate the Track Machines on ……. Railway.

Item Description	Last Done/ Attended	Next Due
PME		
Initial/Refresher Course, IRTMTC, ALD		
Initial/Refresher Course, G & SR, ZRTI, ….		

This certificate is valid upto …….

No……

Date……..

Dy.CE/TM Line/……

Annexure 7.4

Proforma of Machine Ready Memo

FORM NO: TMM-1

Zonal Railway Serial No. …….

Station……. Date ……..

Machine Type	- ……	Machine Type	- ……
Machine No.	- ……	Machine No.	- ……
SSE/JE/TM/I	- ……	SSE/JE/TM/I	- ……
Machine ready time	- ……	Machine ready time	- ……

Machine Type	- ……	Machine Type	- ……
Machine No.	- ……	Machine No.	- ……
SSE/JE/TM/I	- ……	SSE/JE/TM/I	- ……
Machine ready time	- ……	Machine ready time	- ……

Use one box for each machine. SSE/JE/TM/I of leading machine shall co-ordinate for making this memo.

To – SM/……..

SSE/JE/P.Way/…….

Annexure 7.5

Proforma of Block Permit

Form No.TMM-2

Zonal Railway…… Serial No. …….

Track Machine block permit

(SSE/JE/TM,SSE/JE/P.Way/Record)

Station…….Date ……Time ……..Hrs…….. Minutes……..

SSE/JE/TM (Machine Type/No.) ……..

No. of Track Machines to enter into block section for working ……

BLOCK DETAILS

On Track "Track Machine(s)" is granted block on ……Road, between ……station & ……..station, from ……Hrs……..Mts. to ……Hrs. …..Mts. i.e. for ……. Hrs.& …… Mts. Entering from ……..station& clearing into ……..station.

Special instructions if any ……

SIGNAL DETAILS

You are hereby authorized to pass Signal No. ……at "ON" position & proceed beyond/up to …….. Signal.

You are hereby authorized to pass IBS/IBH Signal No. ……at "ON" position. (If available)

Block Back/Block Forward No. ……..in words ……..

CAUTION ORDER DETAILS

Km. ……..to Km…… with ……KMPH due to ……

Km. ……..to Km…… with ……KMPH due to ……

Km. ……..to Km…… with ……KMPH due to ……

PILOT – OUT DETAILS

Your Track Machine (Type of Machine) ……is being piloted-out from(Line)……to ……..(Line). All points enroute are correctly set and padlocked.

If not applicable, may be strike out.

Signature of SSE/JE/TM Signature of SSE/JE/P.Way Signature of Station Master

Time …….. Time ……..(Station Master's stamp)

Annexure 7.6

Instructions for Machine Working in Automatic Signalling Section

1.0 Double Line – In terms of GR – 9.02 - Duties of SSE/JE/TM when Automatic Stop Signal on double line is to be passed at "ON"

(a) When SSE/JE/TM finds an Automatic Stop Signal with an 'A' marker at 'ON', he shall bring his track machine to a stop in the rear of the signal. After bringing his track machine to a stop in the rear of the signal, the SSE/JE/TM shall wait there for **one minute by day** and **two minutes by night**. If after waiting for this period, the signal continues to remain at 'ON', proceed ahead, as far as the line is clear, towards the next stop signal in advance exercising the great caution so as to stop short of any obstruction.

(b) SSE/JE/TM shall show a stop hand signal towards the rear when the track machine has been so stopped at an Automatic Stop signal.

(c) Where owing to the curvature of the line, fog, rain or dust storm or other cause, the line ahead cannot be seen clearly, SSE/JE/TM shall proceed at a very slow speed, which shall under no circumstances exceed 8 kmph. Under these circumstances SSE/JE/TM may seek the assistance of SSE/JE/P.Way.

(d) When so, SSE/JE/P.Way shall accompany SSE/JE/TM in the driving cab, before track machine moves forward, to assist the SSE/JE/TM in keeping a sharp lookout.

(e) When an automatic stop signal has been passed at 'ON' the SSE/JE/TM shall proceed with great caution until the next stop signal is reached. Even if this signal is 'OFF' the SSE/JE/TM shall continue to look out for any possible obstruction short of the same. He shall proceed cautiously up to that signal and shall act upon its indication only after he has reached it.

2.0 Single Line –In terms of G R No. 9.07 – Duties of SSE/JE/TM when an Automatic Stop signal on single line is to be passed at "ON".

(a) When SSE/JE/TM finds an automatic stop signal with an 'A' marker at 'ON' he shall bring his train to a stop in rear of that signal and wait there for one minute by day and two minutes by night.

(b) If after waiting for this period the signal continues to remain at 'ON" and if telephone communication is provided near the signal, SSE/JE/TM shall contact the Station Master of the next block station or the Centralized Traffic Control Operator of the section where Centralized Traffic Control is provided, and obtain his instructions. The Station Master or the Centralized Traffic Control Operator, as the case may be, shall after ascertaining that there is no train ahead up to the next signal and that it is otherwise safe for the SSE/JE/TM to proceed so far as is known, give permission to the SSE/JE/TM to pass the signal in the "ON' position and proceed up to the next signal, as may be provided under special instructions.

(c) If no telephone is provided near the signal or if the telephone communication provided near the signal is out of order and cannot be made use of, the SSE/JE/TM shall give the prescribed code of whistle and exchange

signal with the SSE/JE/TM available on last track machine and then proceed to pass the signal as far as line is clear, up to the next stop signal in advance, exercising great caution so as to stop short of any obstruction.

(d) SSE/JE/TM shall show a stop hand signal towards the rear when the train has been so stopped at an automatic signal.

(e) Where owing to the curvature of the line, fog, rain or dust storm or other cause, the line ahead cannot be seen clearly, SSE/JE/TM shall proceed at a very slow speed, which shall under no circumstances exceed 8 kmph. Under these circumstances SSE/JE/TM may seek the assistance of SSE/JE/P.Way.

(f) When so, SSE/JE/P.Way shall accompany with SSE/JE/TM in the driving cab, before track machine moves forward, to assist the SSE/JE/TM in keeping a sharp look-out.

(g) When an automatic stop signal has been passed at 'ON' the SSE/JE/TM shall proceed with great caution until the next stop signal is reached. Even if this signal is 'OFF', the SSE/JE/TM shall continue to look out for any possible obstruction short of the same. He shall proceed cautiously up to that signal and shall act upon its indication only after he has reached it.

3.0 Procedure to Pass Semi-Automatic Stop Signal at 'ON' – In terms of G & SR NO. 9.14 – Procedure when Semi-Automatic Stop signal is 'ON'.

(a) When a semi-automatic stop signal is worked as an automatic stop signal, rule 9.02 or 9.07 shall apply as the case may be.

(b) When a semi-automatic stop signal is worked as a manual stop signal and becomes defective, it may only be passed under relevant rules.

(c) When SSE/JE/TM is authorised to pass a semi-automatic stop signal at 'ON' by taking 'OFF' the calling-on signal fixed below it, he shall follow the precautions stipulated in Rule 9.02 or 9.07 as the case may be.

4.0 Procedure to Pass Gate Stop Signal at 'ON' In Automatic Signalling Territory – In terms of G & SR No. 9.15 - If SSE/JE/TM finds a Gate stop signal at 'ON' in an Automatic signalling territory –

(a) SSE/JE/TM shall comply with the provisions of rule 9.02 or 9.07 as the case may be, if the 'A' marker is illuminated, or

(b) If the 'A' marker light is extinguished, SSE/JE/TM shall sound the prescribed code of whistle to warn the gateman and bring his track machine to a stop in rear of the signal. If after waiting for one minute by day and two minutes by night, the signal is not taken 'OFF', SSE/JE/TM shall draw his track machine ahead cautiously and stop in the rear of the level crossing. After ascertaining that the gates are closed against road traffic and on getting hand signals from the gateman, and in his absence, SSE/JE/TM shall sound the prescribed code of whistle and cautiously proceed up to the next stop signal complying with the provisions of rules 9.02 or 9.07 as the case may be.

5.0 Procedure for Movement of Track Machines against the Direction of Traffic in Automatic Block System – In terms of G & SR No. 9.13, movement of track machines against the direction of traffic on the automatic block system – in automatic signalling territory, track machine shall run in established direction of traffic only.

Movement of track machines against the established direction of traffic is not permitted. When in an emergency it becomes unavoidably necessary to move track machine against the established direction of traffic, this shall be done only under special instructions, which shall ensure that the line behind the said track machine up to the station in rear is clear and free from obstruction.

6.0 Protection of a Track Machine stopped in an Automatic Block Signalling Section – In terms of G & SR NO. 9.10 - Protection of a track machine stopped in an Automatic Block Signalling Section

(a) When a track machine is stopped in an automatic block signalling section, SSE/JE/TM shall immediately exhibit a stop hand signal towards the rear and check up that the tailboard or taillight is correctly exhibited.

(b) If the stoppage is on account of accident, failure, or obstruction and the machine cannot proceed, SSE/JE/TM shall sound the prescribed code of whistle and track machine shall be protected immediately as per Rule 6.03 except that for the protection of the occupied line one detonator shall be placed at 90 m from the track machine on the way out and similarly two detonators, 10 m part, not less than 180 m from the track machine or at such distance as has been fixed by special instructions.

7.0 Procedure of Block Working of Track Machine in An Automatic Block Signalling Section

(a) In terms of G & SR 4.65-5 (iii, a) – Track machine(s) shall work under the line block only.

(b) Automatic block signalling section shall be treated as absolute block section.

8.0 Intermediate Block Hut/Section

(a) When track machine is going on line clear

(i) While going on line clear from one station to another, if IBS/IBH is found signal at "ON" position then track machine shall be stopped. After 05 minutes SSE/JE/TM shall contact the SM in rear station through the phone provided under the IBH/IBS. If SM allows the machine to go ahead at on position then SM shall give his private No. Thereafter track machine shall proceed to the station in advance.

(ii) While going on line clear from one station to another, if IBS/IBH is found signal at "ON" position then track machine shall be stopped. After 05 minutes SSE/JE/TM shall contact the SM in rear station through the phone provided under the IBH/IBS and if phone provided under the IBS/IBH is found defective and if visibility is clear then track machine shall proceed to the station in advance with 15 kmph, if visibility is not clear then track machine shall proceed to the station in advance with 08 kmph.

(iii) After reaching to the station in advance, SSE/JE/TM shall report as above in writing to SM on duty of station in advance.

(b) When track machine is going for block working IBH/IBS shall be treated as cancelled and track machine may pass IBS/IBH signal at ON position. This fact shall be mentioned in the memo given by station master.

Annexure 7.7

Infringement to Indian Railways Schedule of Dimension by Various Track Machines

(A) Infringement to Schedule of Dimensions – 1939

SN	Name of the Machine	Make	Transport-ation Code	Infringement Description	Actual Dimension of Machine (mm)	Dimension as per SOD (mm)	Speed in kmph	
							Self pro-pelled	In train formation
1.	Continuous Tamping Machine (09 - 32 CSM)	M/s Plasser		i) Min. dia of new wheel tread measured at 63.5 mm from wheel gauge face.	730	914	60	40
				ii) Min. rigid wheel base.	1800	1830		
2.	Points & Crossing Tamping Machine (UNIMAT 08-275-3S)	M/s Plasser		i) Min. dia of new wheel tread measured at 63.5 mm from wheel gauge face.	730	914	60	40
				ii) Min. rigid wheel base.	1800	1830		
				iii) Max. distance between any two adjacent axles.	12200	11890		
3.	Dynamic Track Stabilizer (DGS-62N)	M/s Plasser		i) Min. dia of new wheel tread measured at 63.5 mm from wheel gauge face	730	914	60	50
				ii) Min. rigid wheel base for bogie.	1500	1830		
4.	Ballast Cleaning Machine (RM-80)	M/s Plasser		i) Min. dia of new wheel tread measured at 63.5 mm from wheel gauge face.	900	914	40	30
				ii) Max. distance apart of bogie centres.	22200	14785		
				iii) Max. length of bogie or roof.	29360	21340		
				iv) Max. length of under frames over headstocks.	29360	21030		
				v) Max. length over side buffers.	30600	22300		
				vi) Max. distance apart between adjacent axles.	20370	11890		
5.	Ballast Regulator (66-4)	M/s – Kershaw USA		i) Min. dia of new wheel tread measured at 63.5 mm from wheel gauge face. 4-wheeler vehicles.	838.2	914	50	40
				ii) Max. length of body or roof for 4-wheeler vehicles.	9120	8540		
				iii) Max. length over side buffers	10389	9810		

SN	Name of the Machine	Make	Transp-ortation Code	Infringement Description	Actual Dimension of Machine (mm)	Dimension as per SOD (mm)	Speed in kmph	
							Self pro-pelled.	In train formation
6.	Track Relaying Train (P-811-S)	MIs Tamper Corpora-tion, USA		i) Min. dia of new wheel tread measured at 63.5 mm from wheel gauge face.	838.2	914	5	50
				ii) Min. rigid wheel base for bogie truck.	1727.2	1830		
				iii) Max. distance apart between any two adjacent axles	12072.94	11890		
7.	Points & Crossing Cleaning Machine (RM-76 UHR)	M/s Plasser		i) Min. dia. of new wheel tread measured at 63.5 mm from wheel gauge face.	900	914	40	30
				ii) Max. distance apart of bogie centre.	19500	14785		
				iii) Max. length of body or roof for bogie vehicles.	23490	21340		
				iv) Max. length of under frames over headstocks.	23490	21030		
				v) Max. length over side buffers	24730	22300		
				vi) Max. distance apart between two adjacent axles	17670	11890		
8.	Points & Crossing Changing Machine (AMECA-T-28)	M/s –Ameca Italy		i) Wheel Gauge: Maximum Minimum	 1604 1598	 1602 1599	10	-
				ii) Min. dia of new wheel tread measured at 63.5 mm from wheel gauge face	400	914		
9.	Ballast Cleaning Machine (RM80-92-I)	M/s Plasser		i) Min. dia on wheel tread.	900	914	50	40
				ii) Max. distance apart of bogie centre	22200	14785		
				iii) Max. length of body	28300	21340		
				iv) Max. length over head stock	28300	21030		
				v) Max. length over side buffers.	29540	22300		
				vi) Max distance apart between any two adjacent axles.	20370	11890		
10.	Shoulder Ballast Cleaning Machine (FRM-80)	M/s Plasser		i) Min. dia on wheel tread.	900	914	40	30
				ii) Max. distance apart of bogie centres	16000	14785		
				iii) Max. length of body	38200	21340		
				iv) Max. length over head stocks	38200	21030		
				v) Max. length over side buffers.	39470	22300		
				vi) Max distance apart between any two adjacent axles.	14170	11890		
11.	UTV 502 L	M/s –Tamper Carpora-tion USA		i) Min. dia on new wheel tread.	838.2	914	60	40
				ii) Max. rigid wheel base.	8000	6100		
				iiii) Max. length of body for 4-wheel vehicle 3200 mm wide.	11303	8540		
				iv) Max. length over head stocks.	11303	8540		
				v)Max. length over side buffers.	12522	9810		

SN	Name of the Machine	Make	Transp-ortation Code	Infringement Description	Actual Dimension of Machine (mm)	Dimension as per SOD (mm)	Speed in kmph	
							Self pro-pelled.	In train formation
12.	P&C Tamping Machine 08-275 (UNIMAT).	M/s Plasser		i) Min. dia on wheel tread.	730	914	60	40
				ii) Min. distance apart of bogie centres.	11500	11933		
				iii)Min. bogie wheel base.	1500	1830		
13.	Ballast Regulator 56-3	M/s –Kershaw USA		i) Min. dia on wheel tread.	832	914	60	40
				ii) Max. length of body.	11074	8540		
				iii) Max. length over headstocks.	11074	8540		
				iv) Max. length over side buffers.	12344	9810		
14.	Multipurpose Tamping Machine.	M/s Plasser		i). Min. dia on wheel tread measured at 63.5mm from gauge face.	730	914	60	40
				ii) Min. rigid wheel base for bogie truck of any vehicle.	1800	1830		
15.	High Output Tamping Machine (09-3X) 20.0t axle load.	M/s Plasser	CHTM09-3XP1	i). Min. dia on the tread of new carriage wheel, measured at 63.5mm from the wheel gauge face.	730	914	60	60
				ii) Max. distance apart of bogie centres for bogie vehicles 3250 mm wide.	15700	14785		
				iii) Min. rigid wheel base for bogie truck of any vehicle.	1500-Satellite bogie. 1800-Front and Rear bogies			
				iv) Max. Length of underframes over headstocks for bogie vehicles 3250mm wide.	21700	21030		
				v) Max. Length over side buffers for bogie vehicles 3250mm wide.	22940	22300		
16.	Utility Track Vehicle	M/s-BHEL		i)Distance apart for centres of buffers .	1956	1955	50	50
				ii) Max Length of body or roof for 4 wheeler vehicle 3200 mm wide.	10516	8540		
17.	Shoulder Ballast Cleaning Machine (FRM-85F).	M/s Plasser	CSBC85FP	i) Min. Dia on the tread of new carriage or wagon wheel, measured at 63.5 mm from wheel gauge face.	900	914	40	30
				ii) Max. distance apart of bogie centres .	16000	14785		
				iii) Max. Length of body or roof.	38200	21340		
				iv) Max. Length of under frames over head stocks.	38200	21030		
				v) Max. Length over side buffers.	39440	22300		
				vi) Max. distance apart between any two adjacent axles.	14170	11890		

SN	Name of the Machine	Make	Transp-ortation Code	Infringement Description	Actual Dimension of Machine (mm)	Dimension as per SOD (mm)	Speed in kmph	
							Self pro-pelled.	In train formation
18.	Worksite Tamper Model-VPR-02M	M/s -Metex		i). Min. dia of wheel tread measured at 63.5mm from gauge face.	710.325	914	40	40
				ii) Minimum projection for flange of new tyre, measured from tread at 63.5mm from wheel gauge face.	27.675	28.50		
				iii) Max. thickness of flange of tyre ,measured from wheel gauge face at 13 mm from outer edge of flange.	32.31	28.5		
				iv) Max. Length over side buffers for bogie vehicle 3250mm wide.	23550	22300		
19.	Utility Track Vehicle.	M/s- OEPL		i) Max. thickness of flange of tyre ,measured from wheel gauge face at 13 mm from outer edge of flange.	29.4	28.5	60	50
				ii) Max. rigid wheel base.	6500	6100		
				iii) Max Length of body or roof for 4 wheeled vehicle 3200 mm wide.	9000	8540		
				iv) Max. Length of under frames over head stocks.	9000	8540		
				v) Max. Length over side buffers.	10270	9810		

(B) Infringement to Schedule of Dimensions – 2004

SN	Name of the Machine	Make	Transp-ortation Code	Infringement Description	Actual Dimension of Machine (mm)	Dimension as per SOD (mm)	Speed in kmph	
							Self pro-pelled.	In train formation
1	High Output Tamping Machine (09-3X) 18.5t axle load.	M/s Plasser		i). Min. dia on the tread of new carriage or wagon wheel, measured at 63.5mm from the wheel gauge face.	730-for Satellite bogie. 850- for Front and Rear bogies.	914	50	65
				ii) Max. thickness of flange of tyre ,measured from wheel gauge face at 13 mm from outer edge of flange.	28.68	28.50		
				iii)Maximum height above rail level for floor of any unloaded vehicle.	1370	1345		
				iv) Max. distance apart of bogie centres for bogie vehicles .	15700	15241		
				v) Min. rigid wheel base for bogie truck of any vehicle	1500-for Satellite bogie. 1800-for Front and Rear bogies.	1830		
				vi) Max. Length of body or roof for bogie vehicles.	21700	21340		
				vii) Max. Length over side buffers for bogie vehicles.	22940	22300		
2.	High Output Tamping Machine (09-3X) 21.0t axle load.	M/s Plasser	CHTM09-3XP	i). Min. dia on the tread of new carriage or wagon wheel, measured at 63.5mm from the wheel gauge face.	730-for Satellite	914	50	65
				ii) Max. thickness of flange of tyre ,measured from wheel gauge face at 13 mm from outer edge of flange.	29.46	29.4		
				iii) Max. height above rail level for floor of any unloaded vehicle.	1370	1345		
				iv) Distance apart for centres of buffers	1955	1956		
				v.) Max. distance apart of bogie centres for bogie vehicles	15700	14900		
				vi) Min. rigid wheel base for bogie truck of any vehicle.	1500-for Satellite bogie. 1800-for Front and Rear bogies.	1830		
				vii) Max. Length of body or roof for bogie vehicles.	21700	21340		
				viii) Max. Length over side buffers for bogie vehicles.	22940	22300		

SN	Name of the Machine	Make	Transp-ortation Code	Infringement Description	Actual Dimension of Machine (mm)	Dimension as per SOD (mm)	Speed in kmph	
							Self pro-pelled.	In train formation
3.	Utility Track Vehicle with Mobile Crane.	M/s Phooltas		i) Max. thickness of flange of tyre ,measured from wheel gauge face at 13 mm from outer edge of flange.	29.4	28.5	50	60
				ii) Distance apart for centres of buffers.	1956	1955		
				iii) Max. rigid wheel base	8000	6100		
				iv) Max. Length of body or roof for 4 wheeled vehicle.	11900	8540		
				v) Max. Length over side buffers for 4 wheeled vehicle	13170	9810		
4.	Utility Track Vehicle.	M/s BEML	CUTVB	i) Max. thickness of flange of tyre ,measured from wheel gauge face at 13 mm from outer edge of flange.	29.4	28.5	50	50
				ii) Distance apart for centres of buffers.	1956	1955		
				iii) Max. rigid wheel base for four wheeled vehicles.	7000	6100		
				iv) Max Length of body or roof for 4 wheeled vehicle.	10850	8540		
				v) Max. Length over side buffers for 4 wheeled vehicle.	12120	9810		
5.	Dynamic track Stabilizer	M/s BHEL	CTSBH	i) Maximum height above rail level for floor of any unloaded vehicle.	1450	1345	50	50
6.	Ballast Regulating Machine Model-USP 2000 SWS	M/s Plasser	BRM/USP-2000SWS	i) Min. Dia on the tread of new carriage or wagon wheel, measured at 63.5 mm from wheel gauge face.	850	914	50	60
				ii) Max. thickness of flange of tyre ,measured from wheel gauge face at 13 mm from outer edge of flange.	28.64	28.5		
				iii) Maximum height above rail level for floor of any unloaded vehicle.	2100	1345		
				iv) Distance apart for centres of buffers	1955	1956		
7.	Point & Crossing Tamping Machine (UNIMAT 08-475-4S)	M/s Plasser	PCTM/08-475-4S	i) Max. thickness of flange of tyre ,measured from wheel gauge face at 13 mm from outer edge of flange.	29.46	28.50	50	65
				ii) Maximum height above rail level for floor of any unloaded vehicle.	1370	1345		
				iii) Distance apart for centres of buffers.	1955	1956		
				iv) Min. rigid wheel base for bogie truck of any vehicle.	1800	1830		
				v) Max. Length over side buffers for bogie vehicle.	28370	22300		

SN	Name of the Machine	Make	Transp-ortation Code	Infringement Description	Actual Dimension of Machine (mm)	Dimens-ion as per SOD (mm)	Speed in kmph	
							Self pro-pelled.	In train formation
8.	Lifting, Lining ,Levelling and Tamping Machine Model-08-32C DUOMATIC	M/s Plasser		i) Min. Dia on the tread of new carriage or wagon wheel, measured at 63.5 mm from wheel gauge face.	850	914	50	50
				ii) Max. thickness of flange of tyre ,measured from wheel gauge face at 13 mm from outer edge of flange.	28.64	28.50		
				iii) Distance apart for centres of buffers.	1955	1956		
				iv) Min. rigid wheel base for bogie truck of any vehicle.	1800	1830		
9.	Dynamic Track Stabilizer (Model DSP-C8T)	M/s Metex	DTSC8TMX	i) Min. Dia on the tread of new carriage or wagon wheel, measured at 63.5 mm from wheel gauge face.	860	914	50	50
				ii) Maximum height above rail level for floor of any unloaded vehicle.	1635	1345		
10.	Ballast Regulating Machine Model-PBR 400R	M/s Plasser	CBRM400RP	i) Min. Dia on the tread of new carriage or wagon wheel, measured at 63.5 mm from wheel gauge face.	730	914	50	50
				ii) Max. thickness of flange of tyre ,measured from wheel gauge face at 13 mm from outer edge of flange.	28.64	28.50		
				iii) Distance apart for centres of buffers.	1955	1956		
				iv) Max Length of body or roof for 4 wheeled vehicle.	9750	8540		
				v) Max. Length over side buffers for 4 wheeled vehicle.	11020	9810		
11.	Rail Grinding Machine(RGI-72)	M/s Loram	CRG 72L	i) Maximum height above rail level for floor of any unloaded vehicle.	1683	1345	50	65
12.	Ballast Regulating MachineModel-RPB-01	M/s Metex	CBRM/RPB-01MX	i) Maximum height above rail level for floor of any unloaded vehicle.	1700	1345	50	50
				ii) Distance apart for centres of buffers.	1955	1956		
13.	Ballast Cleaning Machine	M/s BHEL		i) Maximum height above rail level for floor of any unloaded vehicle.	1600	1345	50	50
				ii) Max. distance apart of bogie centres for bogie vehicles.	18000	15241		
				iii) Max. Length of body or roof for bogie vehicles	23840	21340		
				iv) Max. Length over side buffers for bogie vehicles.	25110	22300		
				v) Max. distance apart between any two adjacent axles	15900	12345		
14.	Worksite Tamper without flat car Model-VPR-02M	M/s Metex	TTM/VPR-02M	i) Min. Dia on the tread of new carriage or wagon wheel, measured at 63.5 mm from wheel gauge face.	732	914	50	50

SN	**Name of the Machine**	**Make**	**Transp-ortation Code**	**Infringement Description**	**Actual Dimension of Machine (mm)**	**Dimens-ion as per SOD (mm)**	**Speed in kmph**	
							Self pro-pelled.	**In train formation**
15.	Ballast Regulating Machine Model SPZ 210k	M/s Gemac China	CBRMG	i) Max. rigid wheel base for four wheeled vehicles.	7500	6100	50	60
				ii) Max. length of body or roof for 4 wheeled vehicle.	13590	8540		
				iii) Max. Length over side buffers for 4 wheeled vehicle.	14780	9810		
16.	Multipurpose Tamping Machine Model-Unimat Split Head MFI	M/s Plasser	CMTUSHP	i)Min. Dia on the tread of new carriage or wagon wheel, measured at 63.5 mm from wheel gauge face.	730	914	45	45
				ii)Distance apart from centres of buffers.	1955	1956		
				iii) Max. Length over side buffers for bogie vehicles.	28570	22300		
17.	Rail Borne Maintenance Vehicle (RBMV) Model-RBMV.01	M/s Phooltas	RBMVP	i)Min. Dia on the tread of new carriage or wagon wheel, measured at 63.5 mm from wheel gauge face	952	914	65	65
				ii) Distance apart from centres of buffers .	1956	1956		
				iii)Max. Length over side buffers for bogie vehicles	21337	21340		